Designing For Epistemological Pluralism

A Technical and Ethical Foundation for an Indigenous Values-Aligned Language Model

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Abstract

This paper proposes a framework for designing an AI system that reasons using Indigenous logics instead of Western defaults. It argues that existing large language models reflect narrow ways of thinking, but the solution lies not in neutralizing them. Existing methods focus only on eliminating bias in AI systems; this paper proposes that by doing so, the opportunity to build a model that reasons in radical ways is lost. It makes a case for "re-biasing" a model, to bring a new way of thinking to problems and work towards an integration of multiple different epistemologies in the real world. Instead of building a model immediately, the paper outlines a slower but more ethical approach: scaffolding. This includes collecting community-reviewed testimonies on decision-making, tagging reasoning patterns, building consent protocols, and developing new evaluation methods. The aim is to eventually create an LLM aligned with relational, land-based, and intergenerational thinking, beginning with governance-first design. While preliminary, the paper sketches a path toward building AI systems that support epistemological plurality, especially in fields like sustainability and policy.

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1. Why AI Needs New Foundations

AI is quickly being incorporated into nearly every part of our lives - in governance, education, design, and much more. It has the ability to reason at previously unheard-of levels, but the kind of reasoning it performs represents only the smallest sliver of how human beings actually think.

The core reasoning logic of most current large language models prioritize things like efficiency, cost-benefit, and individual optimization - values that come directly from WEIRD (White, Educated, Industrialized, Rich, Democratic) societies. This isn't necessarily the biggest problem. In reality, for the majority of people using AI for simple, daily tasks, it doesn't pose much of a problem at all. What this paper is concerned with is not that these models are biased or that we need to "de-bias" them to make them neutral, because that work is already being done in dozens of labs. This paper suggests that there is actually a more serious problem at play - that we are not taking advantage of how much we can bias these systems *on purpose.*

If AI is going to help shape our future, narrowing its worldview to one way of thinking limits creativity. When we rely on these LLMs the bigger problem is we're actually unable to be as innovative - they naturally give flat answers and suggestions that are limited to one field of thinking.

This project proposes we actually "rebias" an LLM to fully realize the possibilities present in a different type of thinking: Indigenous reasoning. Indigenous reasoning is vastly different from Western reasoning, with ideas like relationality, land kinship, and seven-generation responsibility, and it's being left behind in the dust. Simply preserving Indigenous reasoning as a cultural artifact and not letting it shape how our systems are built is irresponsible and a disservice not only to Indigenous Peoples, but the rest of the world. To truly push for innovation in AI we must ask what it looks like to build AI systems that reason differently.

This doesn't mean jumping in to build a tool that actually does so yet; the first step is to create the conditions for a system like this to begin to be possible. This paper proposes not a product, but a roadmap and scaffolding for a system that uses Indigenous reasoning systems in its very core. What is actually meant by "scaffolding" is this: the set of conditions - values, protocols, responsibilities, etc. - that guide whether and how such systems should ever be built. It includes a shared understanding of Indigenous reasoning systems, ethical boundaries, and protocols for implementation, adaptation and refusal on the part of communities. This roadmap outlines what must be in place before systems can responsibly reflect Indigenous thinking. We have a responsibility to not only include but *utilize* differing perspectives, in order to unite in an increasingly divisive society, but we have to be extremely careful in approaching the process with humility. It's *incredibly* important to understand the difference between working with Indigenous communities to develop a tool that benefits all involved, and being extractive - simply using valuable knowledge to further innovation for Western communities. This is the reason scaffolding is extremely important; without scaffolding, extractiveness[1] becomes inevitable and we end up perpetuating the colonization of Indigenous knowledge that has existed for centuries.

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[1] Basically, when researchers take sacred Indigenous knowledge to use in studies without having an agreement with the communities.

Another vital concept to this paper is that of ownership. Ownership refers to who has authority over how knowledge is used, shared, modified, or withheld - in this case, within any future AI system. This includes governance, consent, stewardship, and (most importantly) the right of refusal. In Indigenous contexts, ownership may not mean individual possession - it may mean collective custodianship instead, but the point is that these values must shape every part of our AI process.

This paper lays out the early architecture to what an Indigenous-aligned reasoning system could do and look like, as well as what must be in place to build it responsibly. Before that, we'll clarify the actual difference in Indigenous knowledge and Western knowledge, as well as why this is important.

2. Epistemic Foundations & Why Indigenous Reasoning

Indigenous peoples across the world maintain diverse/different knowledge systems deeply rooted in their own unique cultures and histories. However, there are commonalities in Indigenous (as a general term) epistemic foundations that set them apart from Western paradigms. In contrast to Western science's focus on the pursuit of objective and universal truths, Indigenous ways of knowing emphasize context, relationships, and a more holistic understanding of the world. Some key characteristics of these epistemological foundations are:

- **Relationality:** Knowledge is inherently relational, arising from the network of relationships among people, the natural environment, and the spiritual world. Rather than viewing knowledge as something isolated, Indigenous traditions understand all beings and elements as interrelated, with interdependencies between humans, other living creatures, and non-living things.
- **Holism:** Indigenous epistemologies tend to be holistic, meaning they integrate multiple dimensions of experience into one whole. They see the physical, emotional, spiritual, and intellectual aspects of life as inseparable. Understanding any phenomenon requires seeing it in the context of the whole and recognizing that all aspects of existence are connected.
- **Oral Transmission:** Indigenous knowledge is frequently preserved and transmitted through oral traditions (storytelling, song, ceremony, etc.) rather than through written texts. Elders/knowledge-keepers play a vital role in this process, basically acting as living libraries who preserve knowledge. Oral traditions form "the means by which knowledge is reproduced, preserved and conveyed from generation to generation,". A big idea is that knowledge is a collective enterprise: it is validated through group memory and consensus (for example, multiple Elders may "peer review" a story's details for accuracy) and is imbued with the values and histories of the community.
- **Experiential and Place-Based Learning:** Indigenous ways of knowing highly value knowledge gained through direct experience, practice, and observation of the natural world. This approach acknowledges personal experience as a credible source of insight, which is not necessarily true of Western thinking. Additionally, Indigenous knowledge is

> place-based: tied to specific landscapes, ecosystems, and local contexts, making it unique to the needs and challenges of that community and its environment. Basically, context matters a lot; knowledge isn't assumed to be universal but instead related to specific places and communities.

These epistemic foundations[2] form the basis of Indigenous ways of knowing and being. They guide how knowledge is generated, validated, passed on, etc. Cree scholar Willie Ermine observed that Indigenous philosophies come from "a worldview of interrelationships among the spiritual, the natural and the self."[3] These principles have enabled Indigenous societies to sustain their cultures and adapt over millennia, developing ways of understanding that are very distinct from (but not objectively inferior or superior to) Western knowledge systems.

Why Indigenous reasoning specifically?

A really important question that gets brought up is this: "why not Feminist AI? Or Daoist AI? Or Pet AI? Don't they provide different viewpoints too?" While there are thousands of worldviews that exist separate from Indigenous knowledge and Western knowledge, this project focuses on Indigenous reasoning systems because they offer something structurally different. These systems are grounded in relationships to land and rely on non-linear, relational logic, like we talked about above. They are not simply "alternative values layered onto existing AI"; they represent different architectures of reasoning. Especially now, in a time where ecological and social breakdown is a real concern, we need a system rooted in reciprocity to counterbalance (as opposed to neutralize) our current, more extractive way of thinking.

In early tests, I prompted GPT-4 to simulate Indigenous perspectives – for example, asking it to design a sixth-grade science curriculum as if it were an Indigenous community leader. The result, below, emphasized land-based learning, seasonal patterns, and storytelling as a way of knowing. While basic and in need of improvement (this was obviously with no direct Indigenous training data), it shows the potential for a new type of thinking in a real model. If even simulated alignment can shift the structure of outputs, an LLM trained with actual Indigenous logic and testimony could surface entirely new ways of thinking.

This project sees Indigenous reasoning not as a viewpoint to include, but as a system to use and learn from. The goal is to build scaffolding that makes these logics legible and usable, to allow them to guide how reasoning systems are built.

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[2] Blackstock, C. (2007). *The breath of life versus the embodiment of life: Indigenous knowledge and western research*. .

[3] https://opentextbc.ca/indigenizationcurriculumdevelopers/chapter/indigenous-epistemologies-and-pedagogies/

1. Core Cultural Principles (framed in English)

Principle	How it looks in class
Right Practice (Tikanga)	Begin and end each lesson with a short class affirmation that recognises the "life-force" of water and sets expectations for safe, respectful work.
Strong Relationships (Whanaungatanga)	Pupils stay in steady "family groups" for the whole unit, share gear, and keep a joint lab journal. They also connect with a local elder or environmental ranger.
Guardianship (Kaitiakitanga)	Every activity circles back to the question: "How does this help us care for water now and for future generations?" The unit ends with a practical service project at a nearby stream.

Note: Māori terms appear in brackets once, then we use the English equivalent.

2. Big Questions

1. Where does our local water come from and where does it go?
2. How does understanding water science help us protect its health?
3. What actions can young people take to be true guardians of water?

3. Learning Goals

By the end of five weeks, students will:

1. Model particle motion in ice, liquid water and steam, and link these changes to simple cultural stories about land, sea and sky.
2. Explain surface tension, cohesion and adhesion, and relate them to the saying "Water is life for everything."
3. Draw and label the local water cycle, blending scientific terms with traditional place names where appropriate.
4. Test pH, turbidity, temperature and nitrate at two points on the same creek, then compare the numbers with cultural indicators such as "Can we harvest food here?"
5. Design, build and evaluate a low-cost gravity filter; compare its performance with charcoal made from native hardwood.
6. Prepare and deliver a community talk or poster that argues—using evidence—one concrete water-care action the class will take.

A sample 6th-grade water unit designed by GPT-4 (above), emphasizing guardianship

Why is all of this important?
In fields like environmental management and sustainability especially, the value of Indigenous knowledge is increasingly being recognized. Indigenous communities' traditional ecological knowledge, which has been refined over generations, offers crucial insights for solving modern environmental challenges. The Intergovernmental Panel on Climate Change (IPCC) has said that Indigenous peoples' holistic worldview and knowledge of their lands are "a major resource for adapting to climate change". Unlike more top-down approaches, Indigenous knowledge approaches the problem with an understanding of local climate patterns and resilience strategies, all coming from experience. Additionally, major studies agree that traditional Indigenous knowledge is important for biodiversity conservation, as it embodies sustainable practices and deep understanding of local species and habitats. Far from being "outdated," Indigenous knowledge is highly relevant to contemporary issues.

There are also significant efforts underway to integrate Indigenous knowledge through policy and law. The UN *Declaration on the Rights of Indigenous Peoples* (UNDRIP) affirms that Indigenous peoples have the right to "maintain, control, protect and develop"[4] their traditional knowledge and cultural heritage, and this is showing up more and more in practice. For instance, New Zealand has worked to incorporate Māori epistemology into its governance of natural resources. In 2017, the NZ government granted legal personhood to the Whanganui River, a move which emerged from decades of Māori advocacy; this represents a shift toward legal systems that actually incorporate Indigenous worldviews in practice. Similarly, legislatures in other countries (such as India, Colombia, and Ecuador) have started to value Indigenous knowledge in environmental decision-making.

These developments, alongside initiatives to involve Indigenous communities in research and policy-making, demonstrate an increase in this idea called **epistemological pluralism:** the idea that multiple knowledge systems (Indigenous and Western are the focus here, but can include any) can coexist and actually help one another in addressing today's challenges.

In summary, there is an increasing recognition of the wisdom embedded in Indigenous traditions, but ensuring true partnership and respect for Indigenous ways of knowing is (and will keep being) an ongoing process. Nonetheless, this inclusion of Indigenous knowledge signals a broader understanding of knowledge, one that hopefully benefits all of humanity + Earth itself.

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[4] More details on the nitty-gritty of the declaration available (web.law.duke.edu.) Another relevant quote here is "Indigenous people argue that they have legitimate rights to control, access and utilize in any way, including restricting others' access to, knowledge or information that derives from unique cultural histories, expressions, practices and contexts. Indigenous people are looking to intellectual property law as a means to secure these ends." (1.1.2)

3. Scaffolding and Infrastructure

This section outlines the basic framework needed to build towards the reasoning model if it is to be aligned with Indigenous knowledge properly. It isn't a product proposal, but rather a roadmap for what needs to exist before the modeling work can happen, and then how the building will look. The aim is to move slowly, building out the structures needed to make such a model possible in the future.

3.1 Testimony Collection

The starting point is collecting about a few hundred conversations with Indigenous knowledge-holders. Each conversation would focus on a real decision – for example, a choice about land use, conflict, or long-term responsibility. Instead of being purely theoretical and abstract, the interviews are meant to show moments where reasoning naturally becomes visible.

3.2 Reasoning Labels

To make the content useful for a model, excerpts will be labeled with reasoning descriptors, or tags, things like "long-term thinking," "kinship-based reasoning," etc. Each excerpt will get 6-8 labels. Basically, the goal is to make these patterns of reasoning easier to study and reference without reducing or changing them to fit more Western categories. Ideally, these labels would be reviewed and approved by one or two people who understand the worldview the reasoning comes from. In the long term, this forms a kind of index/corpus, to identify patterns.

3.3 How This Leads to a Model

As we know, this scaffolding is meant to support, over time, the development of a language model grounded in Indigenous ways of thinking, which won't necessarily happen right away. What gets built first is the foundation:

- A tagged corpus of decision logic that is reviewed by communities
- Protocols for consent, revision, and knowledge stewardship (see section 4 on guidelines)
- Evals that check whether model outputs reflect the logic

If successful, this early work will show that it's even possible to train a model that reasons differently, and that such a model can be built in collaboration with (not at the expense of) the communities it draws from. Eventually, this tagged corpus would become the training set for a small LLM prototype.

3.4 Eval Framework

Traditional AI benchmarks are likely not as useful here, although once developing we can evaluate this claim better. Outputs will be evaluated on their alignment with the logic they claim to reflect[5]. Early evaluation scaffolding includes:

- Asking original contributors: "Does this response reflect your way of thinking?"
- Testing on new (not included in the testimony corpus) situations: "Can the system reason through a similar kind of decision?"
- Publishing example outputs with context and asking for review

The goal is not really to be objectively correct, but more to get an idea from communities to understand how true to the reasoning the model is.

4. Ethical Guidelines + Guardrails

While there is no "objective correct" in this project, there likely is an "objective wrong"; this project comes with a lot of risk and a responsibility to collect and use data responsibly. The following (provisional) set of basic guidelines[6] is meant to show how to go about the project ethically, and is open to change upon receiving feedback.

Relating to how we interact with communities:

- **Start with relationships as opposed to data**
  Testimonies/data will not be collected before clear relationships are established.
- **Ensure benefit**
  Communities involved in the project will benefit in ways they define, which could include financial compensation, or authorship, or something else entirely.
- **Communal consent**
  Consent from one Indigenous community does not transfer over to a different one. It must be defined in conversation with each community, and it must be ongoing and revocable.

Relating to how we treat the knowledge we collect:

- **Do not flatten difference**
  It's in AI's nature to turn info into easily digestible pieces of data but this risks erasing a lot of complexities. It's necessary to work on how to ensure the data's nuance comes through.

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[5] See the ValuesRAG Cultural Alignment project for the origin of this type of evaluation and how to apply it here

[6] The CARE Principles on Indigenous data governance are also very helpful as a reference point here (https://www.gida-global.org/care)

- **Everything = provisional**
  Participation, protocols, and structures can be revised/withdrawn at any time by communities involved.

Relating to how that knowledge goes into the world:

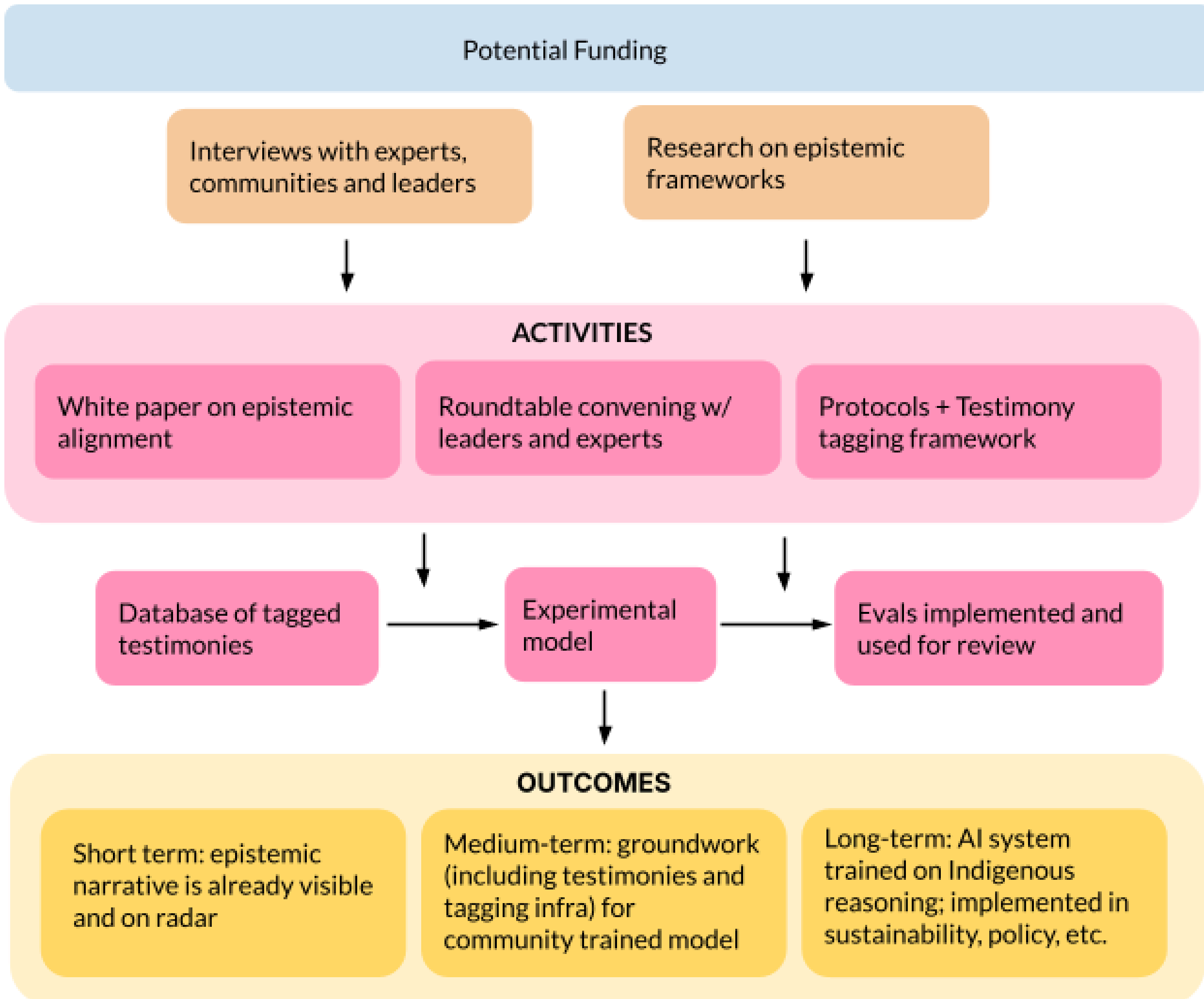
- **Big decisions involve communities**
  Major decisions will be made with a committee of experts and elders.
- **Work in public**
  Drafts of frameworks or technical tools will be shared for feedback before being finalized, and nothing will be released without community review.
- **Value alignment is intentional**
  This project doesn't try to build a neutral system. It is explicitly guided by values like reciprocity, kinship, and circular reasoning. This should be reflected in how the model is designed + evaluated + used.

As stated, the idea is that these guidelines will constantly be changing based on new feedback. The goal is to approach this as ethically as possible while pushing hard for innovation.

5. Theory of Change

As systems theorist Donella Meadows wrote[7], one of the most powerful leverage points to intervene in a system is to shift its underlying mindset and paradigms. This project aims not to simply tweak AI systems but to actually influence the foundational logics they are built on. Below is a bare-bones diagram on the theory of change, condensing the last few sections into just the essentials.

[7] As an aside I do want to elaborate a little more on this theory of leverage points. Meadows says "You could say paradigms are harder to change than anything else about a system, and therefore this item should be lowest on the list, not second-to-highest. But there's nothing physical or expensive or even slow in the process of paradigm change. In a single individual it can happen in a millisecond. All it takes is a click in the mind, a falling of scales from eyes, a new way of seeing. Whole societies are another matter — they resist challenges to their paradigm harder than they resist anything else." The key thing here, in my mind, is that phrase, "a new way of seeing"–however that manifests. At a societal level, as Meadows says, paradigms prove the most difficult to change. I do believe, however, that if one approaches it as changing a mindset on an individual level–to turn the head a few degrees and facilitate this seeing of a just-slightly different world–we gain the power to transform societies.
(Leverage Points: Places to Intervene in a System - The Donella Meadows Project)



Theory of Change Diagram (above) describes the ideal results of this project and outlines the specific actions needed to achieve said outcome (white paper, creation of evals, testimonies, etc.)

6. What Comes Next

As of right now this white paper is still preliminary—proof of what's possible, but in need of more in-depth research, more concrete plans, and the inclusion of everything learned from numerous interviews. Once complete, however, the groundwork will be in place. The next phase of the project is practical: testing what has been proposed. This begins with the Fall 2025 virtual conference/roundtable, which will be the first step toward shared governance, and leads to the testimony-collection period, which will (for this pilot round) probably last 6 months. The roundtable will also provide the opportunity for this white paper to be reviewed, after which it can begin circulating. Early tagging and evaluation protocols will be piloted on a small set of testimonies, and open drafts will be released for review, likely by Summer 2026.

Critically, none of this work assumes finality. Its whole foundation is designed to evolve in public, and with community oversight; the project will continue slowly, guided by feedback.

If this infrastructure works, the long-term vision is to shift how reasoning is represented in AI, including (and especially) in sustainability and policy contexts. The hope is that this infrastructure might inform problems that require more relational ways of reasoning, and eventually help reach true epistemic pluralism.